

Big Data Integration and Storage– Lab 3

Hardik Rathod

8797380

PROG8450

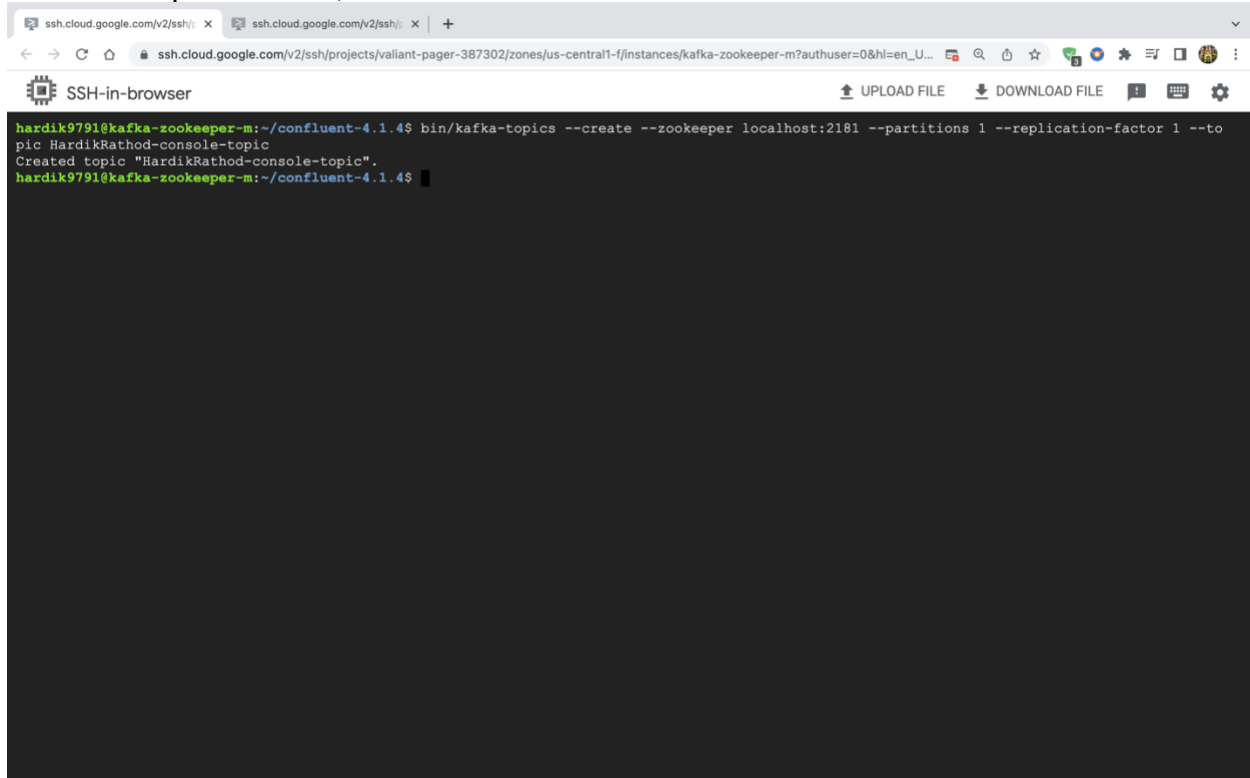
Prof. Saber Amini

July 4th, 2023

We must first install and launch Zookeeper and Kafka in an SSH terminal.

The second step is to create a topic, and in another SSH terminal, my topic is called “HardikRathod-console-topic”.

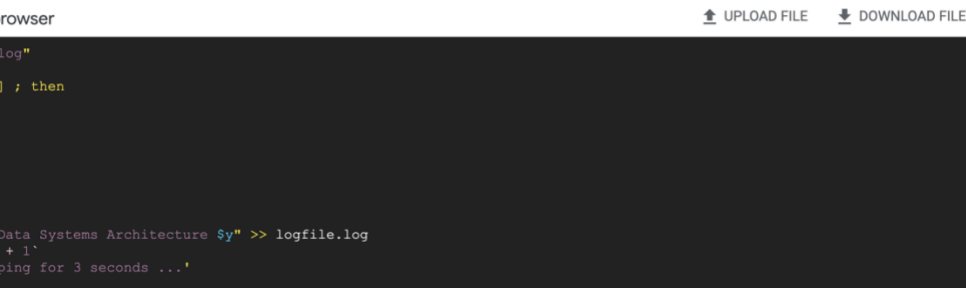
Since the partition in this case is set to 1, the same information will be displayed in both places. If we set the partition to 2, some data will be allocated to Consumer 1 and some to Consumer 2.



The screenshot shows a web browser window with an SSH-in-browser terminal. The terminal output is as follows:

```
hardik9791@kafka-zookeeper-m:~/confluent-4.1.4$ bin/kafka-topics --create --zookeeper localhost:2181 --partitions 1 --replication-factor 1 --topic HardikRathod-console-topic
Created topic "HardikRathod-console-topic".
hardik9791@kafka-zookeeper-m:~/confluent-4.1.4$
```

Afterward, I created `write-to-file.sh`



```
file="logfile.log"

if [ -f $file ] ; then
  rm $file
fi

y=0

while :
do
  echo "Big Data Systems Architecture $y" >> logfile.log
  y=`expr $y + 1`
  echo 'sleeping for 3 seconds ...'
  sleep 3
done
```

I then created the “consumer-one.py” file.

ssh.cloud.google.com/v2/ssh/... | ssh.cloud.google.com/v2/ssh/... | ssh.cloud.google.com/v2/ssh/... | +

ssh.cloud.google.com/v2/ssh/projects/valiant-pager-387302/zones/us-central1-f/instances/kafka-zookeeper-m?authuser=0&hl=en_U...

SSH-in-browser

UPLOAD FILE DOWNLOAD FILE

```
from kafka.consumer import KafkaConsumer

# Kafka consumer configuration
topic = "HardikRathore-console-topic"
brokers = "localhost:9092"

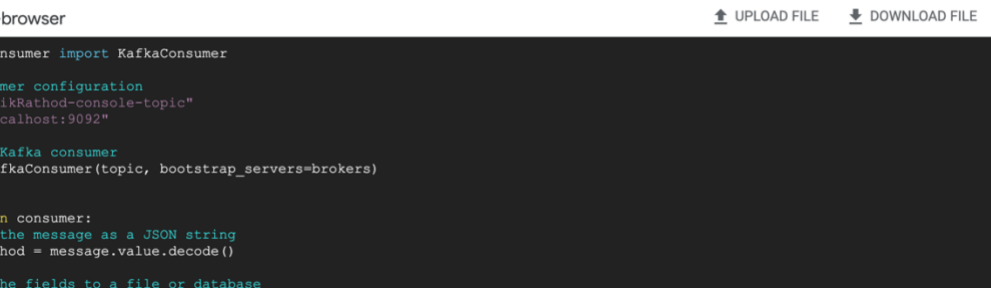
# Create the Kafka consumer
consumer = KafkaConsumer(topic, bootstrap_servers=brokers)

# Continuously poll for new messages
for message in consumer:
    print(message.value.decode())

-- INSERT --
```

4, 22 All

Then created the “consumer-two.py” file.



The screenshot shows a web browser with multiple tabs open, all pointing to 'ssh.cloud.google'. The address bar shows a URL for an SSH connection to a Google Cloud project. The browser window displays a terminal interface with the title 'SSH-in-browser'. The terminal contains Python code for setting up a Kafka consumer. The code imports 'KafkaConsumer' from 'kafka', configures the topic and brokers, creates the consumer, and then iterates over messages, decoding them and writing to a file named 'result.csv'.

```
from kafka.consumer import KafkaConsumer

# Kafka consumer configuration
topic = "HardikRathod-console-topic"
brokers = "localhost:9092"

# Create the Kafka consumer
consumer = KafkaConsumer(topic, bootstrap_servers=brokers)

for message in consumer:
    # Decode the message as a JSON string
    HardikRathod = message.value.decode()

    # Write the fields to a file or database
    with open("result.csv", "a") as f:
        f.write("{}\n".format(HardikRathod))
```

Then, I created “producer-file.py” file.

```
ssh.cloud.google.com/v2/ssh/: X | ssh.cloud.google.com/v2/ssh/: X | +
ssh.cloud.google.com/v2/ssh/projects/valiant-pager-387302/zones/us-central1-f/instances/kafka-zookeeper-m?authuser=0&hl=en_U...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE

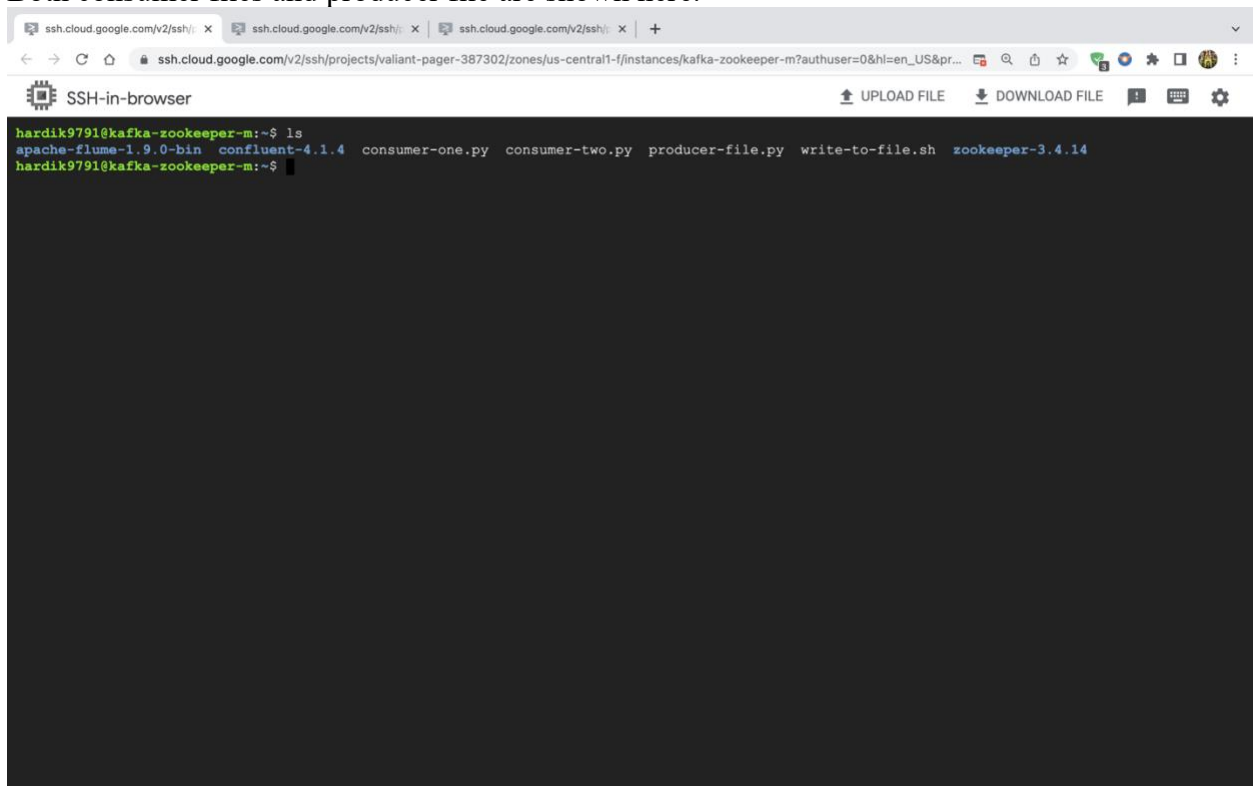
# Kafka producer configuration
topic = "HardikRathod-console-topic"
brokers = "localhost:9092"

# Create the Kafka producer
producer = KafkaProducer(bootstrap_servers=brokers)

# Read the file
filename = "/home/hardik9797/logfile.log"
with open(filename, "r") as file:
    while True:
        # Seek to the end of the file
        file.seek(0, 2)

        # Wait for new data to be written to the file
        while True:
            where = file.tell() # get current position in file
            line = file.readline() # read a line
            if not line: # if there is an empty string (ie no data)
                time.sleep(1) # wait one second
                file.seek(where) # check if there is any new data in the position
            else:
                # there was a string, send new line to the Kafka broker
                producer.send(topic, line.encode())
                producer.flush() # wait until acknowledgement is received from broker before continuing
```

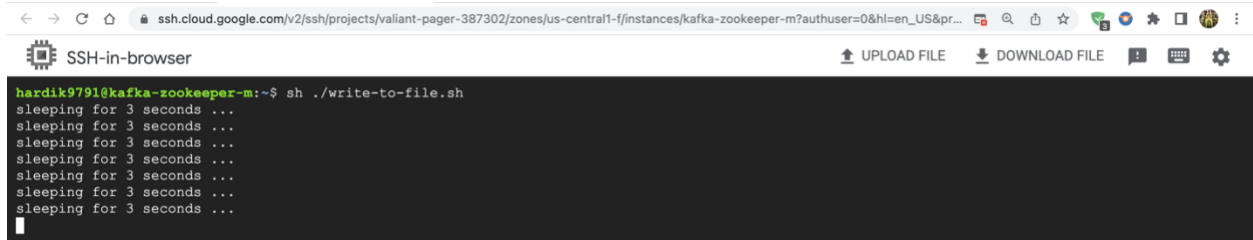
Both consumer files and producer file are shown here.



The screenshot shows a web browser window with multiple tabs. The active tab is titled "SSH-in-browser" and displays a terminal session. The terminal output shows the command `ls` being executed, resulting in a list of files and directories: `apache-flume-1.9.0-bin`, `confluent-4.1.4`, `consumer-one.py`, `consumer-two.py`, `producer-file.py`, `write-to-file.sh`, and `zookeeper-3.4.14`. The terminal prompt is `hardik9791@kafka-zookeeper-m:~$`.

The following steps will demonstrate how I started by first running the write-to-file.sh file, then the consumers and then the producer python files.

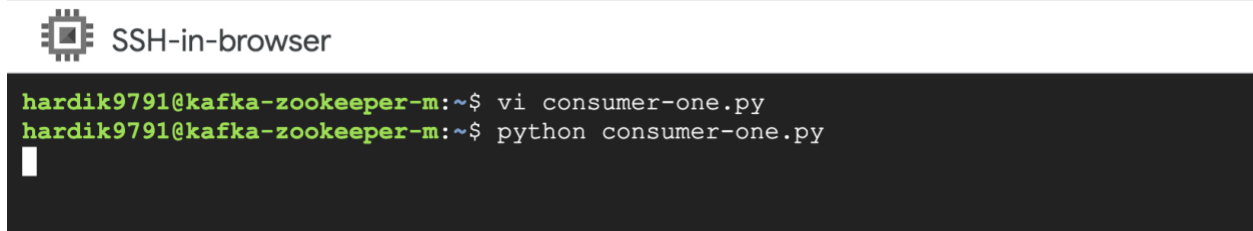
write-to-file.sh



The image shows a terminal window titled "SSH-in-browser" with a URL bar at the top. The terminal output shows a user running a script that repeatedly prints "sleeping for 3 seconds ...".

```
hardik9791@kafka-zookeeper-m:~$ sh ./write-to-file.sh
sleeping for 3 seconds ...
sleeping for 3 seconds ...
sleeping for 3 seconds ...
sleeping for 3 seconds ...
sleeping for 3 seconds ...
sleeping for 3 seconds ...
sleeping for 3 seconds ...
sleeping for 3 seconds ...
```

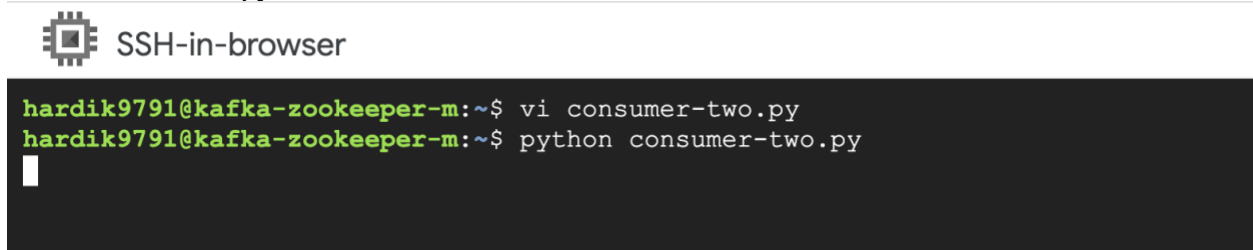
First consumer python file



The image shows a terminal window titled "SSH-in-browser". The terminal output shows a user editing a file with vi and then running it with python.

```
hardik9791@kafka-zookeeper-m:~$ vi consumer-one.py
hardik9791@kafka-zookeeper-m:~$ python consumer-one.py
```

Second consumer python file



The image shows a terminal window titled "SSH-in-browser". The terminal output shows a user editing a file with vi and then running it with python.

```
hardik9791@kafka-zookeeper-m:~$ vi consumer-two.py
hardik9791@kafka-zookeeper-m:~$ python consumer-two.py
```

Producer python file



The image shows a terminal window titled "SSH-in-browser". The terminal output shows a user editing a file with vi and then running it with python.

```
hardik9791@kafka-zookeeper-m:~$ vi producer-file.py
hardik9791@kafka-zookeeper-m:~$ python producer-file.py
```

The producer file has been executed, and the results are starting to show in both consumers files.

First consumer output:



SSH-in-browser

```
hardik9791@kafka-zookeeper-m:~$ python consumer-one.py  
Big Data Systems Architecture 7
```

```
Big Data Systems Architecture 8
```

```
Big Data Systems Architecture 9
```

```
Big Data Systems Architecture 10
```

```
Big Data Systems Architecture 11
```

```
Big Data Systems Architecture 12
```

```
Big Data Systems Architecture 13
```

```
Big Data Systems Architecture 14
```

```
Big Data Systems Architecture 15
```

```
Big Data Systems Architecture 16
```

```
Big Data Systems Architecture 17
```

```
Big Data Systems Architecture 18
```

```
Big Data Systems Architecture 19
```

```
Big Data Systems Architecture 20
```

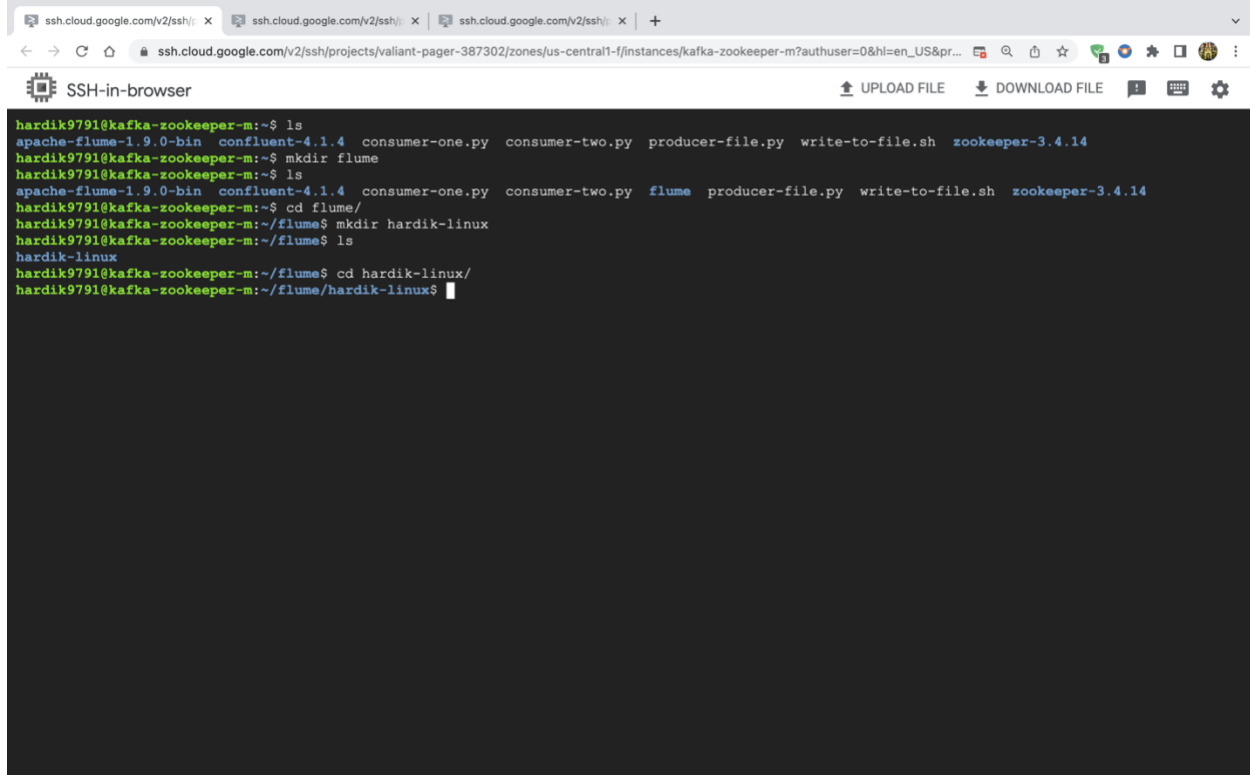
```
Big Data Systems Architecture 21
```

```
□
```

Second consumer output: this output is stored in a csv file

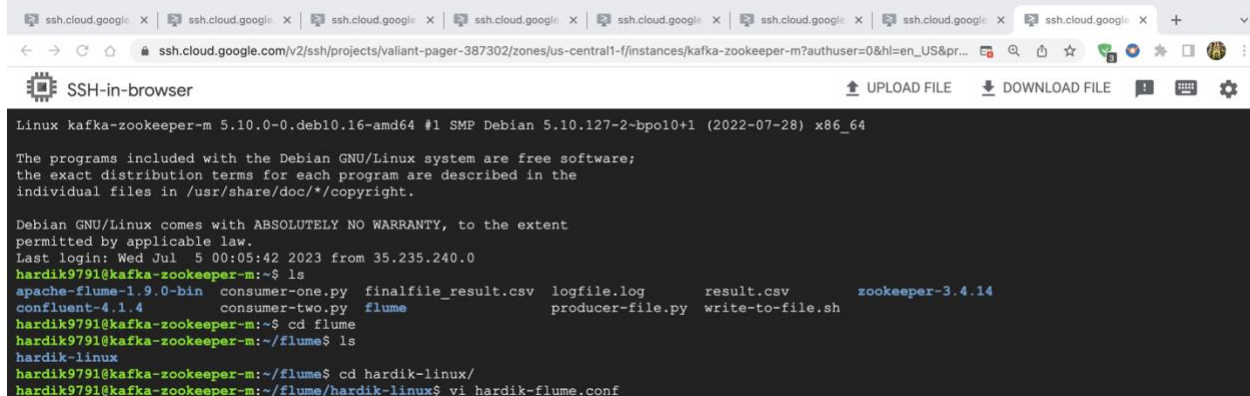


Next, we will install and launch flume then, create a directory called “flume” and in that directory we are going to create another directory called “Hardik-linux” and in that directory we are going to create a config file called “Hardik-flume.conf”.



```
ssh.cloud.google.com/v2/ssh/ X ssh.cloud.google.com/v2/ssh/ X ssh.cloud.google.com/v2/ssh/ X +
ssh.cloud.google.com/v2/ssh/projects/valiant-pager-387302/zones/us-central1-f/instances/kafka-zookeeper-m?authuser=0&hl=en_US&pr...
SSH-in-browser
UPLOAD FILE DOWNLOAD FILE

hardik9791@kafka-zookeeper-m:~$ ls
apache-flume-1.9.0-bin confluent-4.1.4 consumer-one.py consumer-two.py producer-file.py write-to-file.sh zookeeper-3.4.14
hardik9791@kafka-zookeeper-m:~$ mkdir flume
hardik9791@kafka-zookeeper-m:~$ ls
apache-flume-1.9.0-bin confluent-4.1.4 consumer-one.py consumer-two.py flume producer-file.py write-to-file.sh zookeeper-3.4.14
hardik9791@kafka-zookeeper-m:~$ cd flume/
hardik9791@kafka-zookeeper-m:~/flume$ mkdir hardik-linux
hardik9791@kafka-zookeeper-m:~/flume$ ls
hardik-linux
hardik9791@kafka-zookeeper-m:~/flume$ cd hardik-linux/
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$
```



```
ssh.cloud.google.com/v2/ssh/projects/valiant-pager-387302/zones/us-central1-f/instances/kafka-zookeeper-m?authuser=0&hl=en_US&pr...
SSH-in-browser
UPLOAD FILE DOWNLOAD FILE

Linux kafka-zookeeper-m 5.10.0-0.deb10.16-amd64 #1 SMP Debian 5.10.127-2-bpo10+1 (2022-07-28) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jul 5 00:05:42 2023 from 35.235.240.0
hardik9791@kafka-zookeeper-m:~$ ls
apache-flume-1.9.0-bin confluent-4.1.4 consumer-one.py finalfile_result.csv logfile.log result.csv zookeeper-3.4.14
confluent-4.1.4 consumer-two.py flume producer-file.py write-to-file.sh
hardik9791@kafka-zookeeper-m:~$ cd flume
hardik9791@kafka-zookeeper-m:~/flume$ ls
hardik-linux
hardik9791@kafka-zookeeper-m:~/flume$ cd hardik-linux/
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$ vi hardik-flume.conf
```

Below is the code for “hardik-flume.conf” file

[illegible]

I will now launch the Flume Agent, and the command that was used is shown in the screenshot below:

```
Linux kafka-zookeeper-m 5.10.0-0.deb10.16-amd64 #1 SMP Debian 5.10.127-2-bpo10+1 (2022-07-28) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jul 5 00:05:42 2023 from 35.235.240.0
hardik9791@kafka-zookeeper-m:~$ ls
apache-flume-1.9.0-bin  consumer-one.py  finalfile_result.csv  logfile.log  result.csv  zookeeper-3.4.14
confluent-4.1.4        consumer-two.py  flume                producer-file.py  write-to-file.sh
hardik9791@kafka-zookeeper-m:~$ cd flume
hardik9791@kafka-zookeeper-m:~/flume$ ls
hardik-linux
hardik9791@kafka-zookeeper-m:~/flume$ cd hardik-linux/
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$ vi hardik-flume.conf
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$ vi hardik-flume.conf
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$ flume-ng agent --conf /home/hardik9791/flume/hardik-linux/ -f /home/hardik9791/flume/hardik-
linux/hardik-flume.conf -Dflume.root.logger=DEBUG,console -n agent
Info: Including Hadoop libraries found via (/usr/lib/hadoop/bin/hadoop) for HDFS access
/usr/lib/hadoop/libexec/hadoop-functions.sh: line 2365: HADOOP_ORG.APACHE.FLUME.TOOLS.GETJAVAPROPERTY_USER: invalid variable name
/usr/lib/hadoop/libexec/hadoop-functions.sh: line 2460: HADOOP_ORG.APACHE.FLUME.TOOLS.GETJAVAPROPERTY_OPTS: invalid variable name
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Xmx20m -Dflume.root.logger=DEBUG,console -cp '/home/hardik9791/flume/hardik-linux:/home/hardi
k9791/apache-flume-1.9.0-bin/lib/*:/etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/./*:/usr/lib/hadoop-hdfs./usr/lib/hadoop-hdfs/lib
/*:/usr/lib/hadoop-hdfs./usr/lib/hadoop-mapreduce./usr/lib/hadoop-yarn/lib/*:/usr/lib/hadoop-yarn./usr/lib/local/share/google/dataproc/lib/ranger_gcs
plugin_client.jar:/usr/local/share/google/dataproc/lib/spark-metrics-listener-dataproc-2.0-1.5.0.jar:/usr/local/share/google/dataproc/lib/ranger_gcs
-metrics-listener.jar:/lib/*' -Djava.library.path=/usr/lib/hadoop/lib/native org.apache.flume.node.Application -f /home/hardik9791/flume/hardi
k-linux/hardik-flume.conf -n agent
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hardik9791/apache-flume-1.9.0-bin/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class
]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2023-07-05 00:17:10,289 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider starting
2023-07-05 00:17:10,292 INFO node.PollingPropertiesFileConfigurationProvider: Reloading configuration file:/home/hardik9791/flume/hardik-linux/
```

I will demonstrate several hdfs and linux file outputs in this step:

```
ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | +
ssh.cloud.google.com/v2/ssh/projects/valiant-pager-387302/zones/us-central1-f/instances/kafka-zookeeper-m?authuser=0&hl=en_US&pr...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
Linux kafka-zookeeper-m 5.10.0-0.deb10.16-amd64 #1 SMP Debian 5.10.127-2-bpo10+1 (2022-07-28) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jul 5 00:10:17 2023 from 35.235.241.17
hardik9791@kafka-zookeeper-m:~$ hadoop fs -ls /BigDataahardik/flume/hardik-hdfs
Found 19 items
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234517
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234518
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234519
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234520
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234521
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234522
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234523
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234524
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234525
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234526
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234527
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234528
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234529
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234530
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234531
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234532
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234533
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234534
-rw-r--r-- 1 hardik9791 hadoop 0 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234535.hardik
hardik9791@kafka-zookeeper-m:~$
```

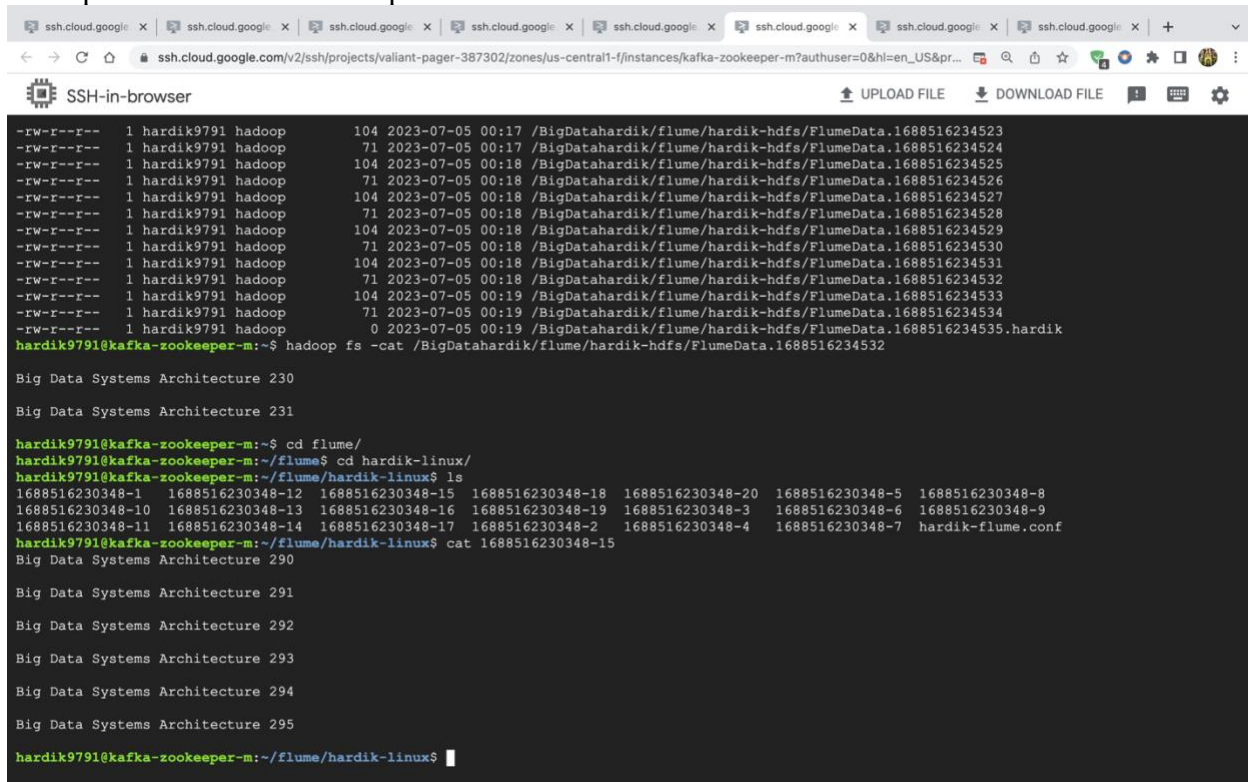
Screenshot of a few HDFS-stored files

```
ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | ssh.cloud.google x | +
ssh.cloud.google.com/v2/ssh/projects/valiant-pager-387302/zones/us-central1-f/instances/kafka-zookeeper-m?authuser=0&hl=en_US&pr...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
Linux kafka-zookeeper-m 5.10.0-0.deb10.16-amd64 #1 SMP Debian 5.10.127-2-bpo10+1 (2022-07-28) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jul 5 00:10:17 2023 from 35.235.241.17
hardik9791@kafka-zookeeper-m:~$ hadoop fs -ls /BigDataahardik/flume/hardik-hdfs
Found 19 items
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234517
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234518
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234519
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234520
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234521
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234522
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234523
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234524
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234525
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234526
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234527
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234528
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234529
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234530
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234531
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234532
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234533
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234534
-rw-r--r-- 1 hardik9791 hadoop 0 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234535.hardik
hardik9791@kafka-zookeeper-m:~$ hadoop fs -cat /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234535
Big Data Systems Architecture 230
Big Data Systems Architecture 231
hardik9791@kafka-zookeeper-m:~$
```

A couple of the linux files outputs are listed below



```
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234523
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:17 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234524
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234525
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234526
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234527
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234528
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234529
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234530
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234531
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:18 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234532
-rw-r--r-- 1 hardik9791 hadoop 104 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234533
-rw-r--r-- 1 hardik9791 hadoop 71 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234534
-rw-r--r-- 1 hardik9791 hadoop 0 2023-07-05 00:19 /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234535.hardik
hardik9791@kafka-zookeeper-m:~$ hadoop fs -cat /BigDataahardik/flume/hardik-hdfs/FlumeData.1688516234532

Big Data Systems Architecture 230

Big Data Systems Architecture 231

hardik9791@kafka-zookeeper-m:~$ cd flume/
hardik9791@kafka-zookeeper-m:~/flume$ cd hardik-linux/
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$ ls
1688516230348-1 1688516230348-12 1688516230348-15 1688516230348-18 1688516230348-20 1688516230348-5 1688516230348-8
1688516230348-10 1688516230348-13 1688516230348-16 1688516230348-19 1688516230348-3 1688516230348-6 1688516230348-9
1688516230348-11 1688516230348-14 1688516230348-17 1688516230348-2 1688516230348-4 1688516230348-7 hardik-flume.conf
hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$ cat 1688516230348-15

Big Data Systems Architecture 290

Big Data Systems Architecture 291

Big Data Systems Architecture 292

Big Data Systems Architecture 293

Big Data Systems Architecture 294

Big Data Systems Architecture 295

hardik9791@kafka-zookeeper-m:~/flume/hardik-linux$
```

Video Recording Link:

https://conestogac.zoom.us/rec/share/7y7_jDhsuJAw6Ej2U4pK6tCq7yj1LQ9hD9OmQojU8vPhiyksGhhbN-lqpOSXaOxi.LUsbRjKRwnliwJEu?startTime=1688526747000